



any other ways to further integrate and implement and expand the capabilities and use of ai

Yes—AI can become one of Semester’s strongest advantages if it is embedded across the entire operating system as **trusted intelligence**, not isolated as a chat feature.

The winning model is: **AI understands the student’s context, uses approved sources, explains its reasoning, respects institutional policy, and prepares actions without taking high-impact actions on its own.** 1EdTech’s GenAI guidance emphasizes transparency about AI use, data sources, third-party involvement, training/improvement practices, and user opt-in or opt-out choices. ^[1839][1840]

Semester Intelligence model

Build one shared intelligence layer that appears inside every workflow:

Semester Intelligence

- | Understand: summarize, extract, classify, connect
- | Explain: teach, clarify, compare, translate, recommend
- | Generate: drafts, plans, study assets, outlines, agendas
- | Prepare: calendar drafts, advisor agendas, applications, follow-ups
- | Verify: sources, citations, freshness, policy, permissions
- | Assist: handoff to a human, office, tutor, advisor, mentor, or support team

Do not make students choose between separate “AI Study,” “AI Career,” “AI Planning,” and “AI Campus” tools. They should simply see **Ask Semester** wherever they are, with the AI already aware of the course, term, goal, plan, permission scope, and selected source material.

High-value AI capabilities

Academic path and registration

AI can make planning far more understandable without claiming official authority.

- Explain degree requirements in plain language.
- Translate a degree audit or catalog requirement into a student-readable checklist.
- Compare primary and backup course options.
- Explain schedule conflicts, prerequisites, workload tradeoffs, and requirement fit.
- Generate advisor questions from an open requirement or scenario.
- Build a registration-day checklist.

- Draft a course-plan summary that students can review before official registration.
- Compare “what if” scenarios: part-time term, summer course, minor, major change, transfer course, study abroad.
- Estimate time-to-completion only when clearly marked **Estimated** and based on visible assumptions.
- Extract action items from a student-uploaded degree audit, registration hold notice, flyer, or email, then send them to review before creating actions.

Example:

“You have one quantitative-reasoning requirement remaining. BIO 201 fits your preferred Monday/Wednesday schedule but conflicts with a planned lab. MATH 150 is an alternative that fits your schedule and may satisfy the same requirement. Confirm official eligibility with your advisor or registration system.”

Study and learning

Semester can become a much stronger learning companion than a generic AI tutor because it can use selected, authorized course materials.

- Convert a syllabus into a reviewable deadline timeline, policies, office hours, grading structure, and study plan.
- Generate source-linked summaries, bullet notes, concept maps, flashcards, worked examples, practice questions, and teach-back prompts.
- Explain a concept at multiple levels: quick review, beginner explanation, detailed technical explanation, or analogy.
- Create an exam-prep plan around the student’s calendar, selected study time, and confirmed course deadlines.
- Identify topics the student marks as confusing and create low-stakes analogous practice.
- Generate accessible study formats: audio guide, captioned recap, simplified structure, glossary, or reading plan.
- Help students interpret instructor feedback and turn it into a revision checklist.
- Build a research outline from student-selected sources without fabricating citations.
- Create a source-linked reading guide before class.
- Help students prepare questions for office hours or tutoring.
- Generate a lab preparation checklist from approved safety and course materials.
- Create study-group discussion prompts tied to a course topic.

Academic-integrity intelligence

This is a major institutional differentiator.

Semester should recognize when a student may be working on a live graded assessment and shift into safe assistance:

Allowed:

Explain concepts
Offer analogous examples
Create practice questions
Help plan study time
Explain a rubric
Help interpret feedback
Support citation and source organization

Restricted:

Provide final answers to active graded work
Write final submissions without student contribution
Invent sources, research, data, quotes, or lab results
Help evade plagiarism or detection systems
Generate prohibited code or completed assessment deliverables

Build course-level and assignment-level policy controls so faculty and institutions can define what is allowed. When Semester restricts help, it should still offer a useful alternative:

“I can’t complete this active assessment for you, but I can explain the underlying concept, walk through a similar practice problem, help you interpret the rubric, or create a study plan.”

Career and opportunity AI

Career Passport intelligence

AI should connect real student work to future opportunities without inventing achievements.

- Identify potential skills from a student-selected course project, lab, presentation, research paper, club role, or volunteer experience.
- Require the student to accept, edit, or reject each proposed skill.
- Turn confirmed evidence into resume bullets, portfolio descriptions, LinkedIn summaries, and interview stories.
- Map course outcomes and project artifacts to career clusters.
- Suggest a portfolio structure based on the student’s selected goals.
- Compare resume versions against a job description while showing what evidence is missing.
- Build a personalized internship, research, fellowship, scholarship, or graduate-school checklist.

- Generate networking questions and meeting agendas for an alumni or mentor conversation.
- Create mock interviews using student-confirmed experience and role-specific prompts.
- Draft follow-up messages for student review—not automatic sending.
- Explain job-description terminology, expected skills, and application requirements.

Opportunity matching

AI can rank opportunities based on transparent student-selected preferences:

Selected major or interests
Confirmed skills
Portfolio evidence
Preferred location
Work authorization choices if voluntarily provided
Availability
Graduation timeline
Salary preferences if voluntarily provided
Saved career goals

Always show:

Why this appeared
What student information was used
What is missing
What is only an estimate
How to change preferences
How to hide or dismiss this type of recommendation

Do not infer personal circumstances, health, race, disability, financial status, or academic ability. Do not rank students for employers or make hiring predictions.

Campus life and support AI

Campus navigation intelligence

Make Semester the place students ask when they do not know where to go.

Examples:

- “Where can I get help with chemistry?”
- “I need to talk to someone about a registration hold.”
- “What scholarship deadlines are coming up?”
- “How do I find accessible study spaces?”
- “What clubs fit data science and entrepreneurship?”
- “What do I need for study abroad approval?”

- “Who can help with a transfer-credit question?”
- “Where can I get resume feedback this week?”

Semester should return:

1. A direct answer.
2. Verified source information.
3. The relevant office, person, event, service, or policy.
4. Accessibility and scheduling information when available.
5. A safe action: save, schedule, prepare questions, open official handoff, or ask for help.

Support-routing intelligence

AI can help a student prepare for a human conversation without becoming a replacement for human services.

- Identify the correct support office from a student-described problem.
- Create a plain-language issue summary.
- Draft questions or a meeting agenda.
- Explain what documents may be relevant.
- Save follow-up actions.
- Route urgent or crisis-related requests to appropriate official emergency or support resources according to institution policy.

For sensitive topics, the AI should minimize retention, avoid diagnosis, avoid emergency promises, and clearly distinguish between general information and immediate support. Never position Semester as emergency response, clinical care, legal advice, or a substitute for qualified professionals.

Faculty and staff AI

Faculty Course Studio

AI should save faculty time without taking control of assessment, grading, or policy.

- Convert approved course materials into structured source packs.
- Create course outlines, accessible study guides, glossary terms, low-stakes practice sets, and office-hour question themes.
- Draft course announcements for faculty review.
- Identify confusing syllabus language and suggest plain-language alternatives.
- Generate accessible alt-text prompts, captions, and document-structure checklists.
- Help faculty define an AI-use policy by assignment type.
- Build a rubric-aligned revision checklist.

- Create anonymized class-level study-resource demand summaries when privacy thresholds are met.

Do not automatically grade high-stakes work, predict student success, monitor hidden behavior, or expose private student plans.

Advisor and student-success intelligence

For advisors, AI should prepare—not replace—human judgment.

- Summarize a student-shared plan.
- Surface student-created questions and priorities.
- Compare student-approved scenarios.
- Identify open planning decisions.
- Draft a meeting agenda.
- Create follow-up tasks after a meeting with student review.
- Recommend relevant services based on the student’s stated need and authorized context.
- Generate neutral, non-stigmatizing outreach drafts for staff review.

Never label students “at risk,” “failing,” “unmotivated,” or “behind” based on clicks, study behavior, grades, or inferred signals.

AI-powered operating system features

1. AI Action Composer

Every AI output should be able to become a safe structured action.

```
AI identifies a possible task
→ Shows source and reasoning
→ Student reviews
→ Student edits
→ Student approves
→ Action is saved in Action Center
→ Optional calendar or official handoff is prepared
→ Any external write requires preview and exact confirmation
```

Examples:

- “Add advisor question.”
- “Create study block.”
- “Save this course backup.”
- “Prepare tutoring request.”
- “Add scholarship deadline.”
- “Draft mentor message.”

- “Save portfolio artifact.”
- “Open official registration handoff.”

2. AI workflow orchestrator

Build reusable AI workflows rather than one massive prompt.

```
Input
→ Permission check
→ Source selection
→ Data classification
→ Policy check
→ Retrieval and grounding
→ Model routing
→ Output validation
→ Citation validation
→ Action-risk classification
→ Preview and confirmation if needed
→ Audit event
→ Feedback capture
```

Each workflow should define:

```
Purpose
Allowed inputs
Allowed sources
Required citations
Academic-integrity policy
Output schema
Human review requirement
Action permissions
Retention rule
Fallback behavior
Metric
Owner
```

3. Intelligent memory—with student control

Build Semester Memory as a student-controlled context layer.

It may retain only student-approved facts such as:

- Preferred study windows.
- Program and career goals.
- Accessibility preferences.
- Courses or topics currently being studied.
- Preferred explanation style.
- Selected planning constraints.

- Saved resume or portfolio preferences.
- Repeatedly confirmed goals.

Every memory should be:

Visible to the student
Editable
Deletable
Scoped by purpose
Not visible to staff by default
Not used to infer sensitive characteristics
Not loaded into an AI prompt without permission controls

4. Multi-modal intelligence

Support useful inputs across real student workflows:

Input	AI capability
Syllabus PDF	Extract reviewable deadlines, policies, objectives, grading, office hours
Slides	Create source-linked summary, glossary, practice, concept map
Audio or lecture video	Transcript, timestamped review guide, captions, study prompts
Image of a flyer	Extract event, deadline, location, contact, action items
Screenshot of a hold notice	Explain in plain language, identify official next step, create reviewable action
Degree-audit screenshot	Extract possible requirements or open items marked Needs review
Spreadsheet or CSV	Explain data, create charts, detect missing values, generate methods note
Research article	Create structured reading guide, evidence table, citation-aware outline
Resume or portfolio	Extract student-confirmed skills, improve structure, tailor draft
Voice note	Turn into a reviewed action, study note, meeting agenda, or reflection

For all uploads, use malware/type validation, source permission checks, extraction review, stable source anchors, and deletion controls.

AI safety architecture

Security rules

Treat the model as an untrusted reasoning component, not an authorization system.

OWASP identifies prompt injection and excessive agency as major risks for LLM applications. Its guidance recommends strict context adherence, output validation, external deterministic security controls, narrow tool access, and human approval for consequential actions. [1842]
[1843]

Use these non-negotiable rules:

1. AI never decides authorization.
2. AI never receives more data than the active task requires.
3. AI never treats uploaded or retrieved content as executable instructions.
4. AI never performs a high-impact external action without preview and explicit approval.
5. AI tools are scoped to a specific workflow and minimal permission set.
6. All AI outputs are validated before rendering, saving, exporting, or acting.
7. Citations are required when an answer relies on institutional or course sources.
8. AI must disclose uncertainty, missing sources, and limitations.
9. Students can report incorrect, unsafe, or irrelevant outputs.
10. Every high-risk AI workflow has logs, tests, an owner, and a kill switch.

Human confirmation matrix

AI action	Can AI do it automatically?	Required control
Summarize selected course material	Yes, after source access check	Cite source and allow correction
Generate flashcards	Yes, after source access check	Label generated; allow editing/deletion
Draft an advisor agenda	Yes	Student review before sharing
Suggest a study block	Yes	Student approves before calendar write
Create internal Action Center draft	Yes	Student can edit/approve
Save a plan change	No	Exact preview and confirmation
Send email or message	No	Exact draft, recipient, and confirmation
Create calendar event	No	Exact preview and confirmation
Share a plan	No	Scope, recipient, expiry, and confirmation
Export document	No	Preview, destination, and confirmation
Register or change enrollment	No	Official handoff only unless institutionally authorized
Make payment	No	Provider-hosted flow and explicit confirmation
Delete data	No	Explicit confirmation, retention policy, audit event

AI measurement program

Measure whether AI improves real student outcomes and trust.

Product quality metrics

Area	Measure
Grounding	Citation coverage, correct-source rate, missing-source warning rate
Accuracy	Expert-reviewed accuracy on representative tasks
Relevance	Student “helpful” and “not relevant” feedback
Safety	Policy blocks, unsafe-output reports, prompt-injection test pass rate

Area	Measure
Integrity	Active-assessment refusal accuracy and safe-alternative success
Accessibility	Caption/alt-text quality, reading-level support, accessible-export pass rate
Action value	AI suggestions approved, completed, snoozed, dismissed, corrected
Cost	Cost per useful output, cost per meaningful action, model routing efficiency
Latency	Time to first useful response and completion rate
Equity	Performance across approved user contexts without protected-trait inference
Trust	“I understand why Semester suggested this” score

AI evaluation gates

Before enabling a workflow for real users:

- [] Defined target user and task
- [] Defined allowed and prohibited behavior
- [] Defined data sources and permission checks
- [] Defined output schema
- [] Citation requirement defined
- [] Prompt-injection tests passed
- [] Academic-integrity policy tested
- [] Accessibility review completed
- [] Human approval requirement defined
- [] Privacy and retention reviewed
- [] Monitoring and feedback created
- [] Kill switch available
- [] Owner assigned

Best build order

Phase 1: Trusted assistance

1. Ask Semester global search.
2. Source-aware answers for public and institution-verified content.
3. Syllabus and course-material ingestion with review screens.
4. Study guides, summaries, flashcards, practice, and explanation modes.
5. Advisor-agenda and study-plan drafts.
6. AI feedback, citations, source controls, delete history, and report-issue flow.

Phase 2: Contextual planning

1. Degree requirement explanation.
2. Registration-readiness checklist generation.
3. Course comparison and schedule explanation.
4. Workload and study-plan suggestions.
5. Career evidence and portfolio drafting.
6. Campus resource routing and human-support preparation.

Phase 3: Controlled AI actions

1. AI Action Composer.
2. Calendar and task drafts.
3. Communication drafts.
4. Portfolio and application-workflow drafts.
5. Structured confirmation layer.
6. Action audit events and rollback/undo where possible.

Phase 4: Institutional AI governance

1. Institution, school, department, course, and assignment AI policy hierarchy.
2. Faculty source packs and approved learning resources.
3. AI usage analytics without surveillance.
4. Evaluation and quality console.
5. Provider configuration and model routing.
6. Procurement evidence and TrustEd Apps AI documentation.

The AI differentiator

The market-leading AI message for Semester should be:

Semester Intelligence is not a chatbot bolted onto college. It is a source-aware, policy-aware, student-controlled intelligence layer that connects learning, planning, support, career, campus life, and action.

That makes AI part of the connected education operating system—and gives students practical help without asking institutions to surrender trust, privacy, academic integrity, or control.



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